Initials
DV
KJ
SB ✓

Marks	1	2	3	4	5	6	Total
	3	4	1	4	2½	4	18.5

Department of Electrical Engineering - IIT Madras

EE1101 - Signals and Systems - Quiz I

8:00 am - 8:50 am — February 15, 2018 — 20 marks

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Roll Number: EE17B130

Write your answers in the space provided, using correct units, and showing all steps on the question book itself. No marks will be given without steps and clear explanations.

1. Is the signal $x[n] = 3exp\left\{\frac{j3\pi\left(n-\frac{1}{2}\right)}{5}\right\}$ periodic? If so, what is its fundamental period? (3)

A signal $e^{j\omega_0 n}$ is periodic if $\frac{\omega_0}{2\pi} = \frac{m}{N}$ where $\frac{m}{N}$ is a rational number and N would be its period.

Given $x[n] = 3\exp\left\{\frac{j(3\pi(n-\frac{1}{2})}{5}\right\}$. So, here $\omega_0 = \frac{3\pi}{5}$

$\Rightarrow \frac{\omega_0}{2\pi} = \frac{3\pi}{5} \times \frac{1}{2\pi} = \frac{3}{10} = \frac{m}{N}$ so, as $\frac{m}{N}$ is rational number, it is periodic with fundamental period $N = 10$. (1) (1)

Is this function an even function? If not, derive its even component.

$x[n] = 3\exp\left\{\frac{j3\pi(n-\frac{1}{2})}{5}\right\} = 3\left(\cos\left(\frac{3\pi}{5}(n-\frac{1}{2})\right) + j\sin\left(\frac{3\pi}{5}(n-\frac{1}{2})\right)\right)$

$x[-n] = 3\left(\cos\left(\frac{3\pi}{5}(-n-\frac{1}{2})\right) + j\sin\left(\frac{3\pi}{5}(-n-\frac{1}{2})\right)\right)$

So, $x[n] \neq x[-n]$.

which implies, it is not an even function.

even component of the function $x_e[n] = \frac{x[n] + x[-n]}{2}$

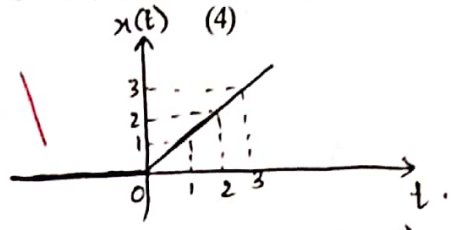
$= \frac{3\left(\cos\frac{3\pi}{5}(n-\frac{1}{2}) + \cos\left\{\frac{3\pi}{5}(-n-\frac{1}{2})\right) + 3j\left(\sin\left(\frac{3\pi}{5}(n-\frac{1}{2})\right) + \sin\frac{3\pi}{5}(-n-\frac{1}{2})\right)}{2}$

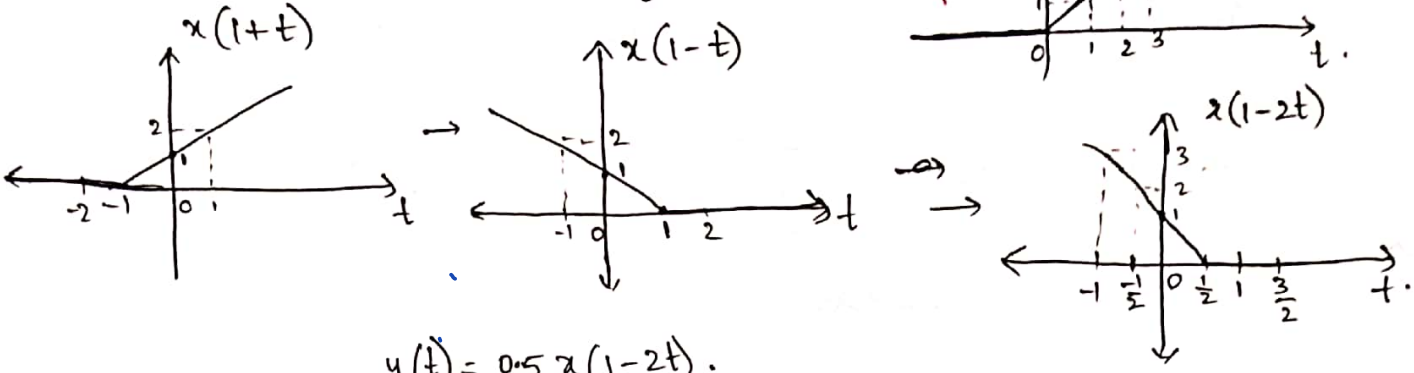
$= \frac{3 \cdot 2 \cdot \cos\frac{3\pi}{5}(\frac{-1}{2})\cos\frac{3\pi}{5}(n) + 3j \cdot 2 \cdot \sin\frac{3\pi}{5}(\frac{-1}{2})\cos\frac{3\pi}{5}(n)}{1}$

$= 3\cos\frac{3\pi}{10}\cos\frac{3\pi}{5}n + 3j\sin\frac{3\pi}{10}\cos\frac{3\pi n}{5} = 3\cos\frac{3\pi n}{5}e^{-j\frac{3\pi}{10}}$ (1)

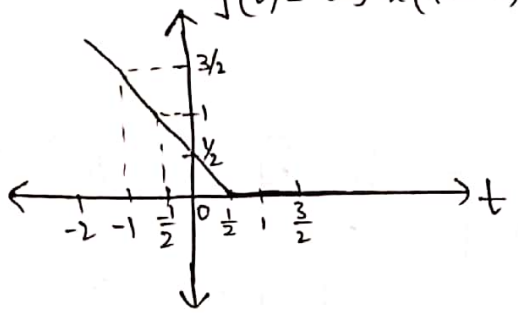
2. If $y(t) = 0.5\,x(1-2t)$ and $x(t) = t\,u(t)$, sketch the following: x(t), y(t) and the derivative of y(t). Mark all critical points. (4)

Given $x(t) = t\,u(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$



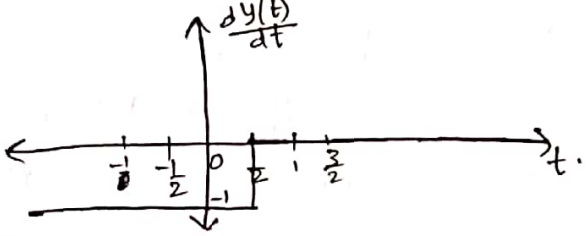


$y(t) = 0.5\,x(1-2t)$.



So, $y(t) = \begin{cases} -t + \frac{1}{2}, & t \leq \frac{1}{2} \\ 0, & t \geq \frac{1}{2} \end{cases}$

$\Rightarrow \frac{dy(t)}{dt} = \begin{cases} -1, & t < \frac{1}{2} \\ 0, & t \geq \frac{1}{2} \end{cases}$



3. Evaluate the following: (2)

a) $\int_{-\infty}^{\infty} \delta(2t+3)e^{-t}dt$

Let $2t+3 = k \Rightarrow t = \frac{k-3}{2} \Rightarrow dt = \frac{dk}{2}$.

$\Rightarrow \int_{-\infty}^{\infty} \delta(2t+3)\,e^{-t}\,dt = \int_{-\infty}^{\infty} \delta(k)\,e^{-\left(\frac{k-3}{2}\right)} \cdot \frac{dk}{2} = \frac{e^{-\left(\frac{0-3}{2}\right)}}{2} \int_{-\infty}^{\infty} \delta(k)\,dk$

$= \frac{e^{3/2}}{2}$ $\left(\because \int_{-\infty}^{\infty} \delta(k)\,dk = 1,\ \int_{-\infty}^{\infty} \phi(k)\,\delta(k)\,dk = \phi(0)\right)$

(b) $\sum_{-\infty}^{\infty} \delta[n-3]\cos\left(\frac{\pi}{2}[n-5]\right)$

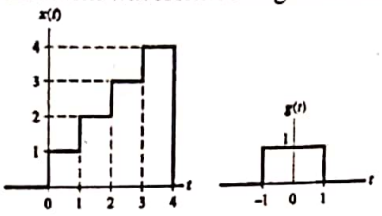
Let $n-3 = K \Rightarrow n = K+3$. then $= \sum_{-\infty}^{\infty} \delta[k]\cos\left(\frac{\pi}{2}[k-2]\right)$

where k is an integer

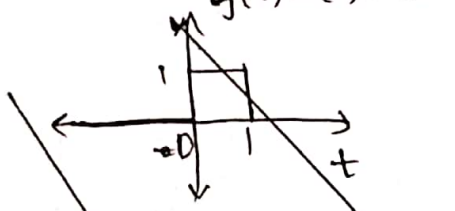
But As k is an integer, k-2 is also an integer.

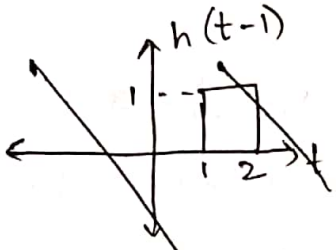
and for

4 Use the rectangular pulse g(t) to construct the waveform x(t) shown in the figures below. Sketch the constituent waveforms and give the final expression for x(t) in terms of g(t). (4)

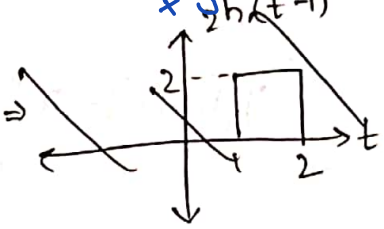


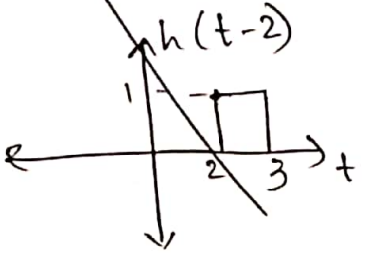
$g(t)u(t) = h(t)$

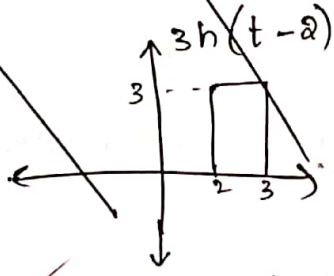


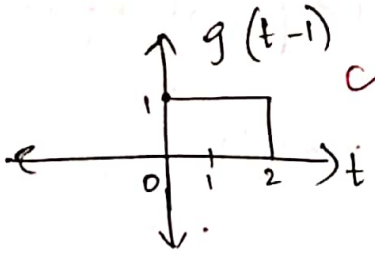


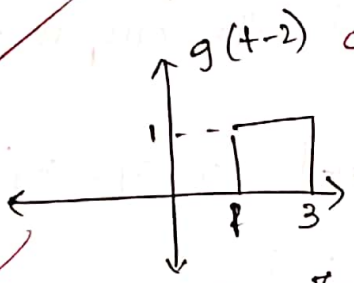
$g(\frac{1}{2}t-1) + g(\frac{2}{3}t-5/3) + g(t-3) + g(2t-7)$

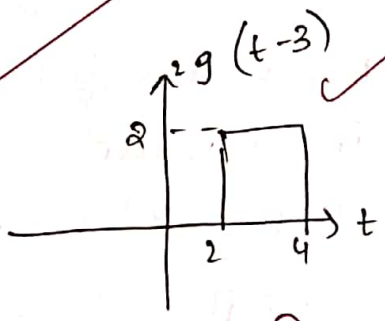




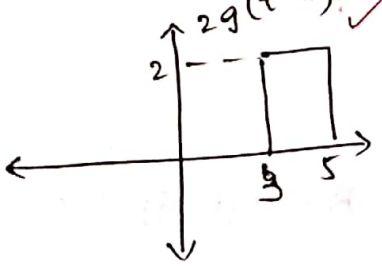


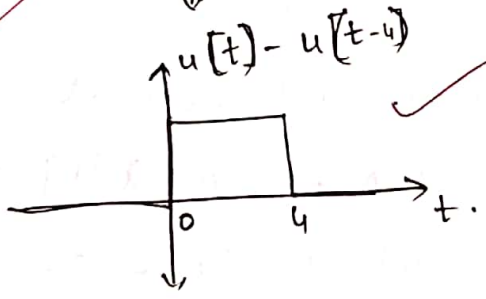




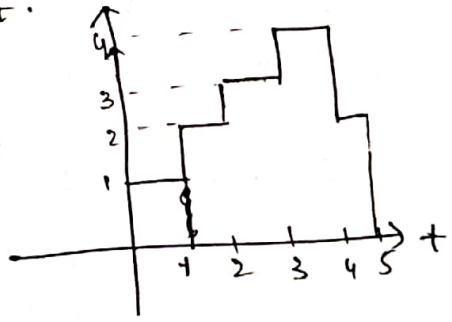


2





So, $g(t-1) + g(t-2) + 2g(t-3) + 2g(t-4) \longrightarrow$



So, $\left(g(t-1)+g(t-2)+2g(t-3)+2g(t-4)\right)\left(u(t)-u(t-4)\right) = x(t)$.

(OR) $x(t) = g(t)u(t) + 2g(t-1)u(t-1) + 3g(t-2)u(t-2) + 4g(t-3)u(t-3)$

2

$\delta[n-3]-\delta[n-1]$

5. The output of a system is $y[n] = x[3-n]-x[n-1]$, where $x[n]$ is the input.

(a) Sketch the impulse response of this system. (3)

$h[n] = \delta[$

0.

(b) Is the system time invariant?

$y[n-n_0] = x[3-n+n_0] - x[n-n_0-1]$.

If the signal is delayed by n_0, $y_1[n] = x[3-n-n_0] - x[n-1-n_0]$

But $y[n-n_0] \neq y_1[n]$.

So, it is not time invariant. (1)

(c) Is the system causal and stable?

For values of $n \leq 1$, the output $y[n]$ depends on future input values.

$y[1] = x[2] - x[0]$, $y[0] = x[3] - x[-1]$

$y[-1] = x[4] - x[-2]$ - - - -

So, the system is not causal. (1/2)

Let $|x[n]| \leq A \Rightarrow -A \leq x[n] \leq A$.

then $-A \leq x[3-n] \leq A$ and also $-A \leq x[n-1] \leq A$.

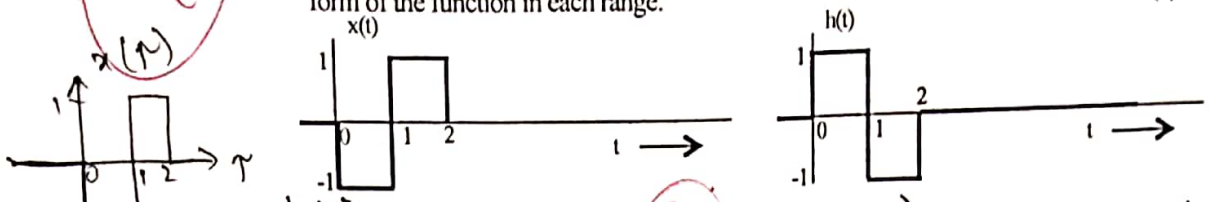
So, As $y[n] = x[3-n] - x[n-1]$

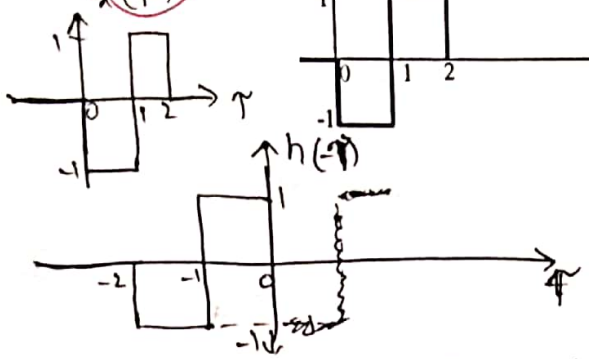
$-2A \leq y[n] \leq 2A$.

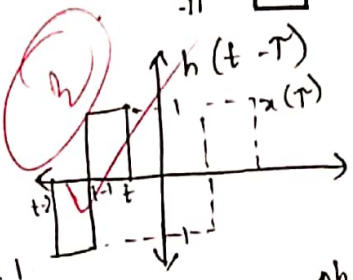
So, for bounded input, we get bounded output.

So, the system is stable. (1)

6. Evaluate and sketch the convolution for a system with impulse response h(t) and input x(t), as shown in the figures below. Clearly mark all critical points and indicate the form of the function in each range. (4)

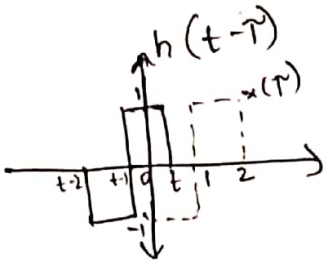






For $t<0$, there is no overlap of $h(t-\tau)$ and $x(\tau)$.

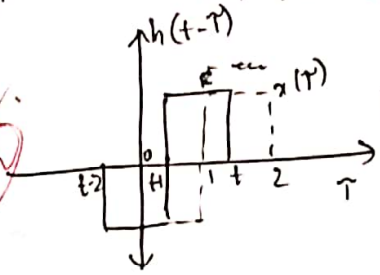
So, $y(t) = 0$



For $0<t<1$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\cdot h(t-\tau)\,d\tau$$

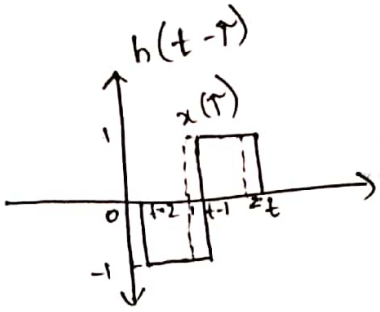
$$= \int_0^t (-1)\cdot 1\cdot d\tau = -t$$



For $1<t<2$

$$y(t) = \int_0^{t-1} (-1)\,dt + \int_{t-1}^{t} (-1)\,dt + \int_{t}^{1} dt$$

$$= t-1+t-1-1+t-1 = 3t-4$$

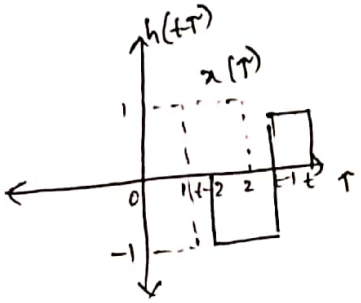


For $2<t<3$

$$y(t) = \int_{t-2}^{1} dt + \int_{1}^{t-1} -1\,dt + \int_{t-1}^{t-2} dt$$

$$= 1-t+2+1-t+t+2-t+1$$

$$= 8-3t$$

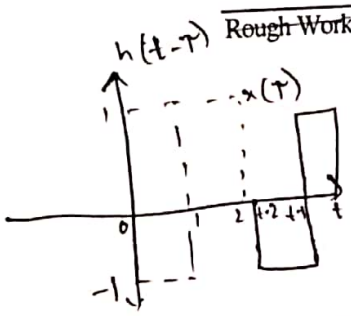


For $3<t<4$

$$y(t) = \int_{t-2}^{2} -1\,dt$$

$$= t-2-2 = t-4.$$

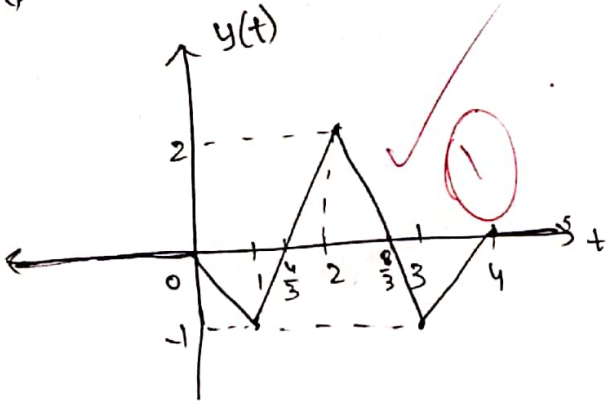
Rough Work



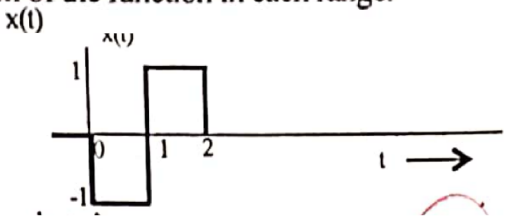
If $t>4$

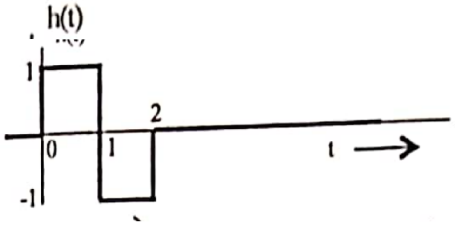
$y(t) = 0$.

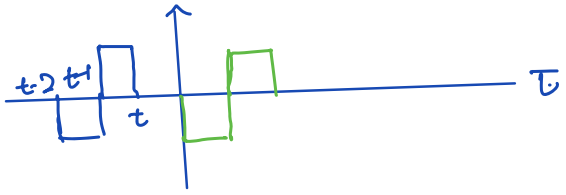
$$y(t) = \begin{cases} 0, & t<0 \\ -t, & 0<t<1 \\ 3t-4, & 1<t<2 \\ 8-3t, & 2<t<3 \\ t-4, & 3<t<4 \\ 0, & t>4 \end{cases}$$



Evaluate and sketch the convolution for a system with impulse response h(t) and input x(t), as shown in the figures below. Clearly mark all critical points and indicate the form of the function in each range. (4)

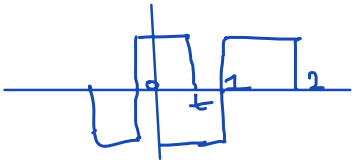






$t<0 \Rightarrow y(t)=0$

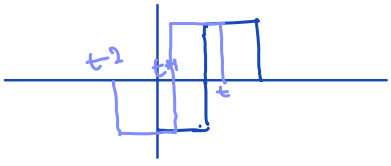
$0\le t\le 1 \Rightarrow$



$$\int_0^t (-1)(1)\,d\tau = -t$$

$$y(t) = -t$$

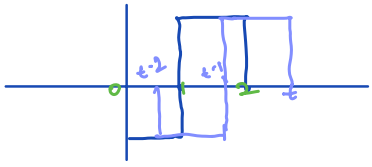
$1\le t\le 2$



$$\int_0^{t-1} (-1)(-1)\,d\tau + \int_{t-1}^{1} (-1)(1)\,d\tau + \int_1^t (1)(1)\,d\tau$$

$$= t-1-1+t-1+t-1 = 3t-4$$

$2\le t\le 3$



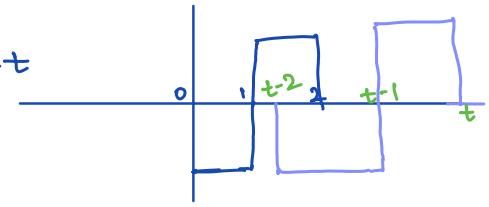
$$\int_{t-2}^{1} (-1)(-1)\,d\tau + \int_1^{t-1} (1)(-1)\,d\tau + \int_{t-1}^{2} (1)(1)\,d\tau$$

$$= 1-(t-2)-t+1+1+3-t$$

$$= -3t+8$$

$3\le t\le 4$

$$\int_{t-2}^{2} (1)(-1)\,d\tau = -4 + t$$



$t \geq 4 \Rightarrow 0$

